

Date 20/01/2027
Class BSc 2nd year
Roll No 68
Shift Day

Physics Practical Sheet
Prithvi Narayan Campus

Experiment No 7
Group P₂
Sub Physics
Set 1

Object of the Experiment (Block Letters) : TO STUDY THE CHARACTERISTICS OF A COMMON EMITTER NPN TRANSISTOR

APPARATUS REQUIRED:

- | | |
|-------------------------|--------------------|
| (i) An n-p-n transistor | (ii) Two battery |
| (iii) Potentiometer | (iv) Voltmeter |
| (v) Micro-ammeter | (vi) Milli-ammeter |
| (vii) Connecting wires | |

THEORY:

A transistor is a semiconductor device mainly used to amplify or switch electronic signals and electrical power. NPN is bipolar junction transistors that are used in amplifiers or oscillators, switches, and other electrical circuits.

Input resistance (R_i) is defined as the resistance offered by the collector junction and has a very small resistance due to forward biasing. Output resistance (R_o) is defined as the resistance offered by the collector junction and has a high value due to reverse biasing. Current gain (β) is defined as the ratio of change in collector current to the change in base current.

Emitter and base are forward biased whereas collector is reverse biased. Base voltage (V_{be}) and base current (I_b) is measured from μA and collector current I_c is denoted by I_c and its voltage is V_{ce} .

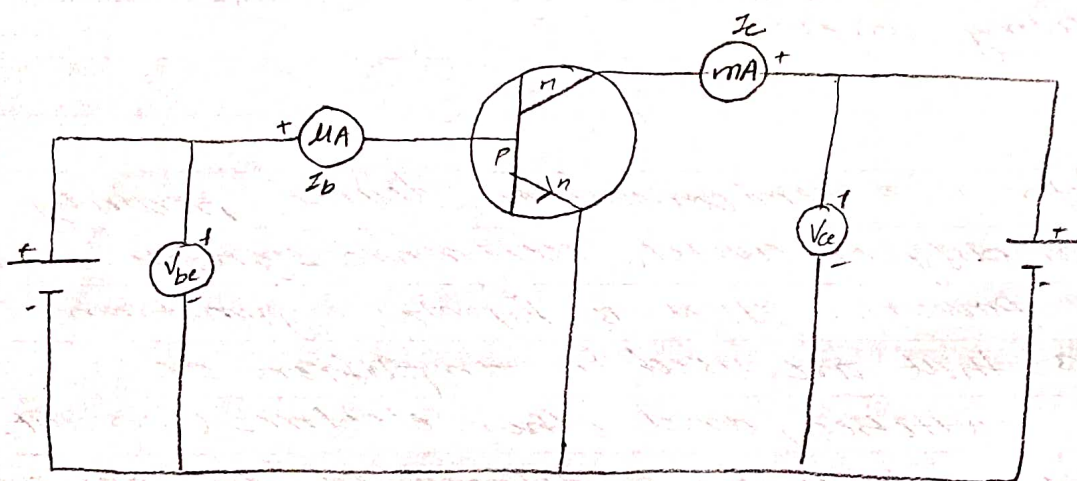


Fig: NPN Transistor CE Characteristics

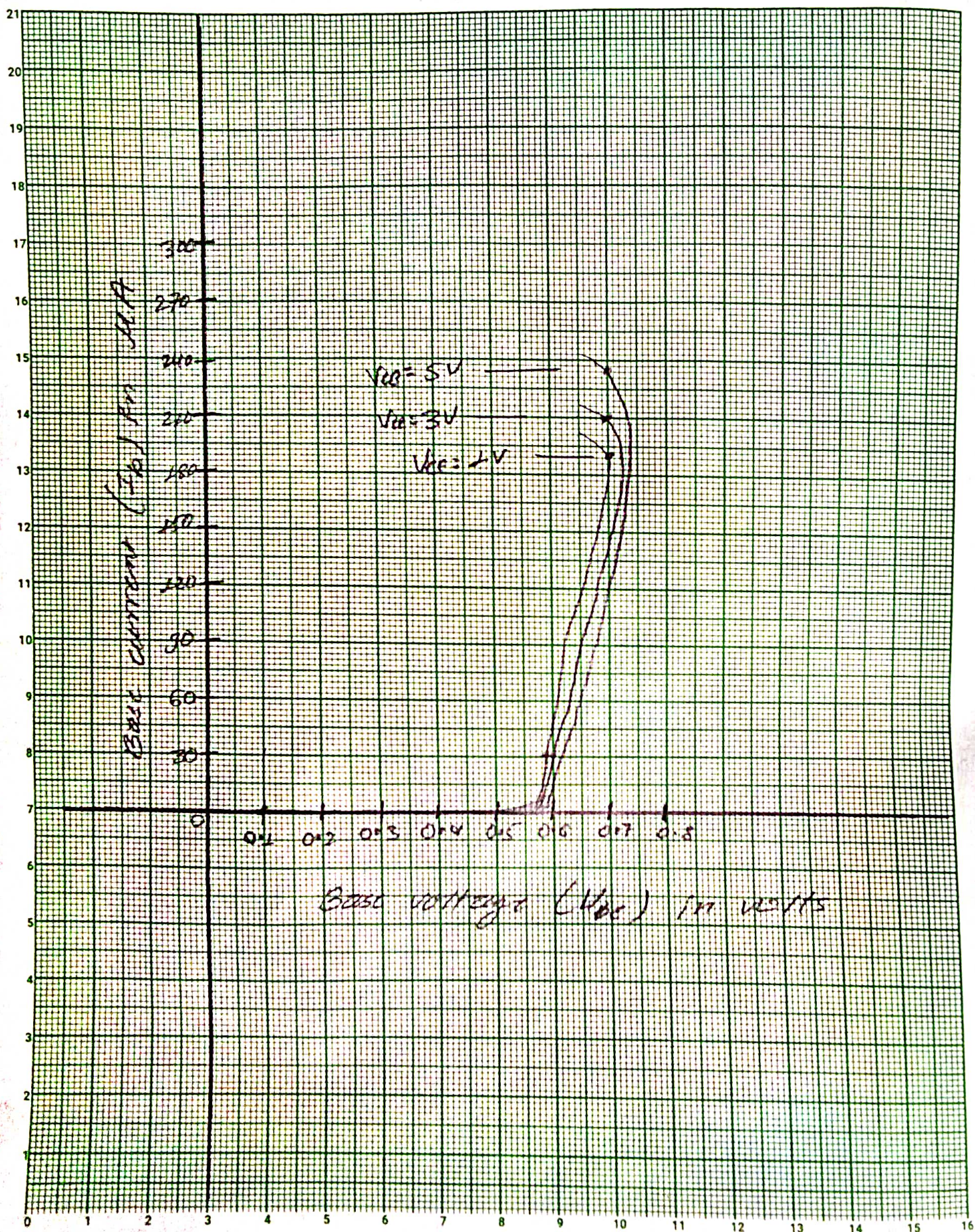
OBSERVATION:

Input Characteristics:

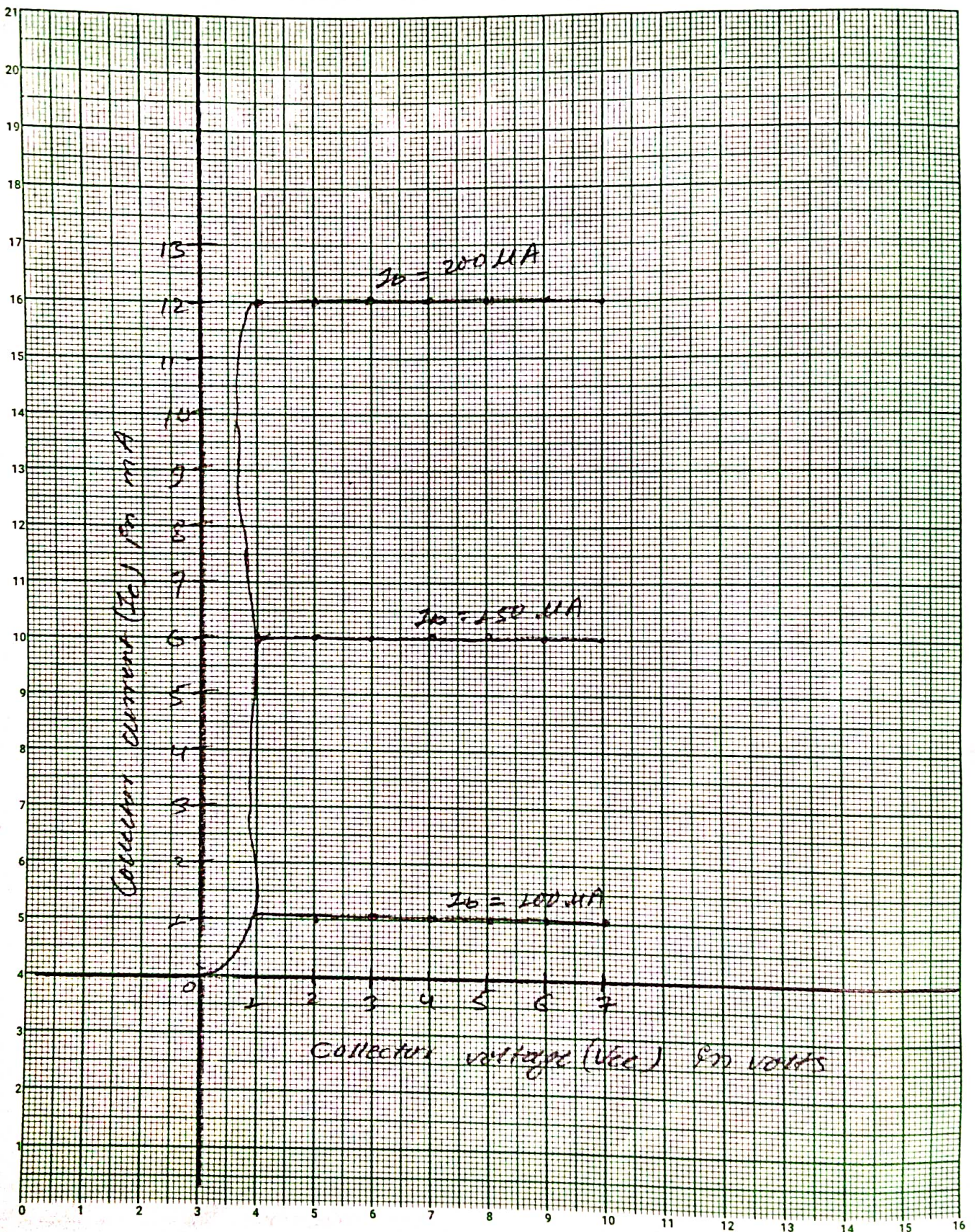
S.N	Base voltage	Base current (I_b) for collector voltage (μA)		
	V_{be} in mV	$V_{ce} = 1 V$	$V_{ce} = 3 V$	$V_{ce} = 5 V$
1.	0	0	0	0
2.	100	0	0	0
3.	200	0	0	0
4.	300	0	0	0
5.	400	0	0	0
6.	500	0	0	0
7.	600	30	29	28
8.	700	190	210	238

Output Characteristics:

S.N	Collector voltage (V_{ce})	Collector current (I_c) for base current (μA)		
	Output in V	$I_b = 100 \mu A$	$I_b = 150 \mu A$	$I_b = 200 \mu A$
1.	0	0	0	0
2.	1	1	6	12
3.	2	1	6	12
4.	3	1	6	12
5.	4	1	6	12
6.	5	1	6	12
7.	6	1	6	12
8.	7	1	6	12



Input characteristics of a common emitter transistor



Output Characteristics of a common emitter transistor

RESULT:

The Input and output characteristics of a common emitter n-p-n transistor was studied.

PRECAUTIONS:

- (i) The connection should be right, clean and tight.
- (ii) Key should be used when the circuit is being used.
- (iii) Beyond breakdown, forward bias voltage should not be applied.
- (iv) Beyond breakdown, reverse bias voltage should not be applied.